

YIFAN JIANG

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EDUCATION

The University of Texas at Austin, Austin, USA

2020 – Present

Ph.D. in Electrical and Computer Engineering¹

- Advisor: [Zhangyang \(Atlas\) Wang](#)
- Committee: [Alan Bovik](#), [Yuke Zhu](#), [Mingyuan Zhou](#), [David Z. Pan](#)

Research Interests: Generative Model, Neural Rendering, Computational Photography

Huazhong University of Science and Technology, Wuhan, China

2015 – 2019

B.E. in Electronic Information Engineering

PUBLICATIONS

(* indicates equal contribution)

- [CVPR] **Y. Jiang**, H. Peter, B. Mildenhall, D. Xu, J. T. Barron, and Z. Wang, “AlignNeRF: High-Fidelity Neural Radiance Fields via Alignment-Aware Training”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023
- [CVPR] D. Xu, **Y. Jiang**, P. Wang, Z. Fan, Y. Wang, and Z. Wang, “NeRF-SOS: Any-View Self-supervised Object Segmentation on Complex Real-World Scenes”, *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023
- [ICLR] Z. Fan, P. Wang, **Y. Jiang**, X. Gong, D. Xu, and Z. Wang, “NeRF-SOS: Any-View Self-supervised Object Segmentation on Complex Real-World Scenes”, *International Conference on Learning Representation*, 2023
- [NeurIPS] D. Xu*, P. Wang*, **Y. Jiang**, Z. Fan, and Z. Wang, “Signal Processing for Implicit Neural Representations”, *Advances in Neural Information Processing Systems*, 2022.
- [ECCV] **Y. Jiang***, D. Xu*, P. Wang, Z. Fan, H. Shi, and Z. Wang. “SinNeRF: Training Neural Radiance Fields on Complex Scenes from a Single Image”, *European Conference on Computer Vision*, 2022.
- [ECCV] **Y. Jiang***, Z. Fan*, P. Wang*, X. Gong, D. Xu, and Z. Wang. “Unified Implicit Neural Stylization”, *European Conference on Computer Vision*, 2022.
- [ECCV] **Y. Jiang**, B. Wronski, B. Mildenhall, J. T. Barron, Z. Wang, and T. Xue. “Fast and High-Quality Image Denoising via Malleable Convolutions”, *European Conference on Computer Vision*, 2022.
- [MM] D. Xu, H. Poghosyan, S. Navasardyan, **Y. Jiang**, H. Shi, and Z. Wang, “ReCoRo: Region-Controllable Robust Light Enhancement by User-Specified Imprecise Masks”, *ACM Multimedia*, 2022
- [WACV] **Y. Jiang**, X. Gong, J. Wu, H. Shi, Z. Yan, and Z. Wang, “AutoX3D: Searching Ultra-Efficient Architecture for Video Understanding”, *IEEE Winter Conference on Applications of Computer Vision*, 2022.
- [TIP] Z. Chen, **Y. Jiang**, D. Liu, and Z. Wang, “CERL: A Unified Optimization Framework for Light Enhancement with Realistic Noise”, *IEEE Transaction on Image Processing*
- [NeurIPS] **Y. Jiang**, S. Chang, and Z. Wang, “TransGAN: Two Pure Transformers can Make One Strong GAN and That Can Scale Up”, *Advances in Neural Information Processing Systems*, 2021.
- [NeurIPS] B. Pan, R. Panda, **Y. Jiang**, Z. Wang, R. Feris, and A. Oliva, “IA-RED2: Interpretability Aware Redundancy Reduction for Vision Transformers”, *Advances in Neural Information Processing Systems*, 2021.
- [ICCV] **Y. Jiang**, H. Zhang, J. Zhang, Y. Wang, Z. Lin, K. Sunkavalli, S. Chen, S. Amirghodsi, S. Kong, and Z. Wang, “SSH: A Self-supervised Framework for Image Harmonization”, *IEEE International Conference on Computer Vision*, 2021.
- [ICLR] T. Meng*, X. Chen*, **Y. Jiang**, and Z. Wang, “A Design Space Study for LISTA and Beyond”, *International Conference on Learning Representations*, 2021.
- [DAC] Y. Fu, Z. Yu, Y. Zhang, **Y. Jiang**, C. Li, Y. Liang, M. Jiang, Z. Wang, and Y. Lin, “InstantNet: Automated Generation and Deployment of Instantaneously Switchable Precision Networks”, *Design Automation Conference*, 2021.
- [TIP] **Y. Jiang**, X. Gong, D. Liu, Y. Cheng, C. Fang, X. Shen, J. Yang, P. Zhou, and Z. Wang, “EnlightenGAN: Deep Light Enhancement without Paired Supervision”, *IEEE Transaction on Image Processing*

¹Studied at Texas A&M University from Aug. 2019 to Aug. 2020; then transferred with my advisor to UT Austin

[ICCV] X. Gong, S. Chang, **Y. Jiang**, and Z. Wang. “AutoGAN: Neural Architecture Search for Generative Adversarial Networks”, *IEEE International Conference on Computer Vision*, 2019.

INTERNSHIP EXPERIENCE

Adobe, San Jose, USA

May. 2022 – Present

Research Intern with [Marc Levoy's Team](#), worked with [Zhihao Xia](#), [Cecilia Zhang](#), [Xiuming Zhang](#) [Jiawen Chen](#).

- Working on monocular video depth estimation

Google Research, Mountain View, USA

May. 2021 – May. 2022

Research Intern with [GCam](#), worked with [Tianfan Xue](#), [Bart Wroński](#), [Ben Mildenhall](#), [Peter Hedman](#), [Jon Barron](#).

- Developed a fast denoising network by predicting spatially-varying kernels at low resolution and using a fast fused op to jointly upsample and apply these kernels at full resolution. The resultant paper was accepted by ECCV'2022
- Designed a high-fidelity neural radiance field that can render high-quality novel view images.

Adobe, San Jose, USA

May. 2020 – Nov. 2020

Research Intern with Applied Research Team (ART), worked with [He Zhang](#) and [Jianming Zhang](#).

- Developed a self-supervised method for image harmonization that does not require human annotation labels. The resultant paper was accepted by ICCV'2021

Bytedance AI Lab, Beijing, China

Jan. 2019 – Aug. 2019

Research Intern with US CV Lab, worked with [Jianchao Yang](#) and [Xiaohui Shen](#) and [Ding Liu](#).

- Designed a jointly image denoising and low-light enhancement algorithm, which was integrated in the selfie camera app [FaceU](#).

MEDIA HIGHLIGHT

- TransGAN was covered by [Quanta Magazine](#) (Mar. 2022) and was highlighted by [Top AI influencers](#) and high-profile [YouTubers](#), as well as considered as [the most influential new paper of the month \(Feb. 2021\)](#).
- AutoGAN was covered by [Synched AI Technology & Industry Review](#) (Aug. 2019), and also featured on [Towards Data Science](#) (Sep. 2019) and [Analytics Magazine](#) (Aug. 2019), etc.

COMMUNITY SERVICES

- Reviewer for: CVPR'2021-2022, ICCV'2021, ECCV'2022, ICML'2022, NeurIPS'2022, ICLR'2023, Siggraph Asia'2022, Siggraph'2022, WACV'2022, Transaction on Image Processing (TIP), International Journal of Computer Vision (IJCV), NeuroComputing, IEEE Robotics and Automation Letters (RA-L)
- Workshop Organizer for: [ECCV RLQ-TOD Workshop 2020](#)

INVITED TALKS

- “Learning to Enhance Low-light Images without Paired Supervision” at [\[IEEE SPS Webinar\]](#), after my TIP work EnlightenGAN was highlighted as one of SPS's top-25 most downloaded articles on IEEE Xplore®, 2021-2022.
- “Fast and High-Quality Image Denoising via Malleable Convolutions” at Adobe, [Marc Levoy's team](#).
- “Vision Transformer for Image Generation, Editing, and Processing” at Google Research, GCam.
- “TransGAN: Two Transformers Can Make One Strong GAN” at [\[cai-workshop\]](#), [\[SHI Lab @University of Oregon\]](#)

AWARDS

- **Apple PhD Scholar in AI/ML** 2023